

OmniMart Market Basket & Cross-Selling Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from over 253,000 orders to optimize cart volumes and cross-selling strategies. The goal is to uncover insights into product affinities (Lead vs. Follower items), habitual purchasing behavior, and discount efficiency to drive Average Order Value (AOV) and protect profit margins.

2. Dataset Summary

- **Raw Transaction Data:** 2.6M rows
- **Raw Product Data:** 92.4K rows
- **Raw Household Demographic Data:** 801 rows
- **Total Orders (Processed):** 253,334
- **Total Households (Tracked):** 5,554,058
- **Key Features:**
 - Cart details (Current AOV, Single-item Basket Rate, Missed Bundle Revenue)
 - Purchase dynamics (Lead Item, Follower Item, Pull Percentages, Times Bought Together)
 - Shopping behavior (Habitual Conversion Rate, Impulse Rate, Customer Lifecycle Weeks)
 - Financials (Total Sales, Total Discount Given, Total Estimated Profit)

3. Data Architecture & Transformation using Snowflake & dbt

We began with data preparation and transformation within our cloud architecture:

- **Data Loading:** Ingested raw transactional data from AWS S3 into Snowflake via Snowpipe.
- **Data Modeling & Transformation:** Used dbt to handle complex transformations, clean data, and build analytical tables.
- **Feature Engineering:**
 - Utilized SQL SELF JOIN with alphabetical deduplication to create Item A and Item B pairings without redundant overlaps.
 - Created Directional Pull metrics to establish "Lead" vs. "Follower" relationships.
 - Calculated Habitual Conversion Rate vs. Impulse Rate using window functions to track purchasing frequency over a 105-week lifecycle.
- **Noise Filtering:** Excluded generic staple goods to ensure association rules highlighted meaningful, actionable cross-selling opportunities.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis to answer key business questions regarding cart optimization:

1. Baseline Cart Metrics - Identified current order values and revenue leakage from unoptimized baskets.

Metric	Value
Current AOV	\$29.11
Single-item Basket Rate	15.3%
Missed Bundle Revenue	\$92,097.78
Revenue Gain Per Conversion	+\$23.73

2. Lead vs. Follower Purchase Drivers - Identified the directional pull between high-volume product pairings to inform discounting strategy.

Lead Item	Follower Item	Pull A to B (%)	Total A Trips	Total B Trips	Times Bought Together
Apples	Baked Bread/Buns/Rolls	53	13,097	60,025	6,902
Bacon	Baked Bread/Buns/Rolls	57	10,730	60,025	6,147
Bag Snacks	Baked Bread/Buns/Rolls	47	41,786	60,025	19,439

3. Discount Efficiency & Financial Impact - Calculated the net financial impact of optimizing discounts based on Lead/Follower behavior.

Metric	Financial Value
Total Sales	\$7,374,424
Total Discount Given	-\$1,362,913

Metric	Financial Value
Total Estimated Profit	\$821,421

4. Habitual vs. Impulse Subscriptions - Identified top product pairs suited for "Subscribe & Save" recurring revenue models.

Item A	Item B	Habitual Conversion Rate (%)	Impulse Rate (%)
Cheese	Salad Dressing/Sandwich Spread	55.91	44.09
Adult Incontinence	Baked Bread/Buns/Rolls	30.00	70.00
Air Care	Bag Snacks	22.00	78.00

5. Dashboard in Tableau

Finally, we built the **OmniMart Command Center** in Tableau to present these insights visually to stakeholders.

- **Cart Optimization View:** Displays AOV, single-item basket rates, and a heatmap of top product pairings by income segment.
- **Profit Drivers View:** Showcases the \$821k estimated profit alongside a detailed breakdown of Lead vs. Follower products.
- **Habitual Targeting View:** Features a scatter plot identifying 5,279 pairs tagged for "Subscribe & Save," alongside a lifecycle line chart proving that habitual household purchases officially overtake impulse buying at **Week 93**.

6. Business Recommendations

- **Margin-Safe Discounting:** Stop heavily discounting "Follower" items (e.g., Bread/Buns when paired with Apples or Bacon). Since customers naturally pull these into their carts, discounts should be reserved strictly for "Lead" items to protect the \$821k in estimated profit.
- **Targeted Bundle Promos:** With 15.3% of orders being single-item baskets, implement checkout prompts that suggest highly correlated items to capture the \$92k in missed bundle revenue.
- **Boost Subscribe & Save:** Launch automated subscription marketing for the 5,279 identified high-habit product pairings (e.g., Cheese & Salad Dressing at a 55.91% habitual rate) to lock in long-term recurring revenue.

- **Lifecycle Marketing:** Focus retention and loyalty marketing efforts heavily before the Week 93 crossover point to successfully convert impulse shoppers into habitual buyers.